INSTRUCTOR AND MATERIAL DEVELOPER GUIDE

Teaching and Learning Framework

A practical guide to teaching concepts clearly, developing effective learning materials, and helping students solve problems independently.

Our approach
Start with a problem. Explain the idea. Build a solution. Improve it. Help students think independently.

CS Made Easy

Prepared by Mikiyas Shegaw, President
Nathan Wondim, Vice President

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1 Purpose and Scope

This guide establishes the common teaching principles and material-development standards followed by CS Made Easy. It is intended for instructors, textbook authors, slide designers, and problem-set developers.

The aim is to ensure that our courses help students understand concepts, apply them meaningfully, and develop the ability to solve unfamiliar problems independently.

The principles apply across our courses. Their implementation may vary according to the subject, learning objectives, and needs of the students.

2 Our Teaching Philosophy

At CS Made Easy, we believe students learn more effectively when they understand why an idea matters, how it works, and how it can be applied.

Our teaching should move beyond presenting definitions and finished solutions. We should show students how problems are approached, how ideas are developed, and why particular solutions work.

Our approach rests on five principles:

1. **Purpose:** Give students a reason to learn a concept by connecting it to a problem or question.
2. **Understanding:** Explain ideas clearly and develop understanding before relying on formal terminology.
3. **Progression:** Introduce concepts in a logical order, building on what students already know.
4. **Application:** Demonstrate how ideas are used through examples, practical activities, and small projects.
5. **Independence:** Prepare students to reason through and solve problems without step-by-step guidance.

3 Problem-Driven Teaching

Whenever appropriate, a lesson should begin with a problem, question, or practical requirement. Students should see what needs to be solved before learning the concept that helps them solve it.

The instructor then guides students through the development of a solution:

1. Present the problem and establish the goal.
2. Develop a simple initial solution.
3. Identify what the solution cannot yet do.
4. Introduce the concept needed to address the limitation.
5. Improve the solution and explain why the change works.
6. Test the result and allow students to practise.

Each step should have a clear purpose. The initial solution should be simple enough to understand, and each improvement should introduce or reinforce a meaningful idea.

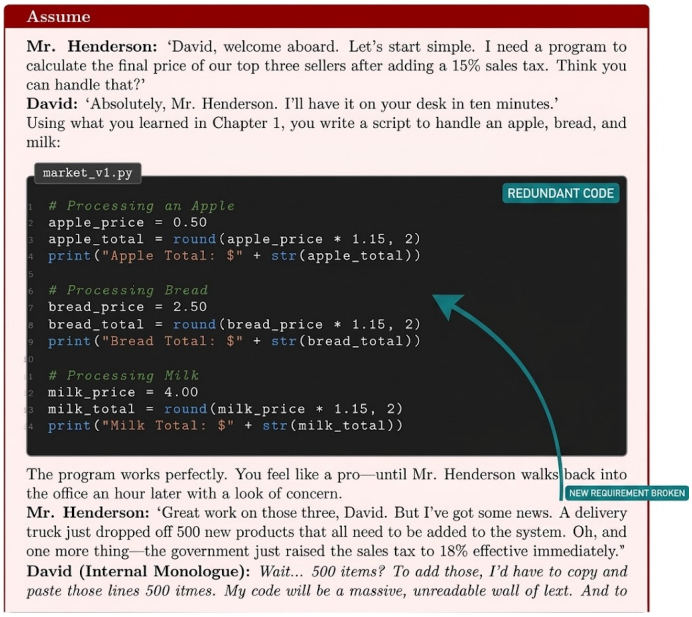


Figure 1: Problem-driven teaching: moving from a problem to a solution through explanation, development, and improvement.

This sequence is a guide, not a rigid requirement. Some subjects and concepts may need a different approach, but students should always understand the purpose of what they are learning.

4 How to Develop Textbook Chapters

A chapter should guide the reader from an initial question or idea toward a clear understanding of the concept and its application. It should teach, not merely summarize information.

Authors should follow these guidelines:

- **Establish a purpose.** Introduce a problem, question, or motivation where appropriate.
- **Explain ideas clearly.** Use accessible language and explain unfamiliar terminology.
- **Build progressively.** Introduce concepts in a logical order and explain the connections between them.
- **Show worked examples.** Explain the reasoning behind each important step rather than presenting only the final result.
- **Connect ideas to practice.** Show how concepts help solve a problem or contribute to a larger solution.
- **Address likely questions.** Include explanations of common confusions, misconceptions, and alternative ways of understanding difficult ideas.
- **Prepare students for independent work.** End with the understanding needed to attempt problems without copying the chapter's examples.

4.1 Using Teacher–Student Dialogue

Dialogue can make a chapter more engaging and help explain ideas in a natural, intuitive way. A teacher may ask a question, respond to a student's confusion, or guide the student toward understanding a concept.

Dialogue should serve a teaching purpose. It should address a genuine question, explain a difficult idea, or reveal the reasoning behind a solution. It should not be added merely for decoration.

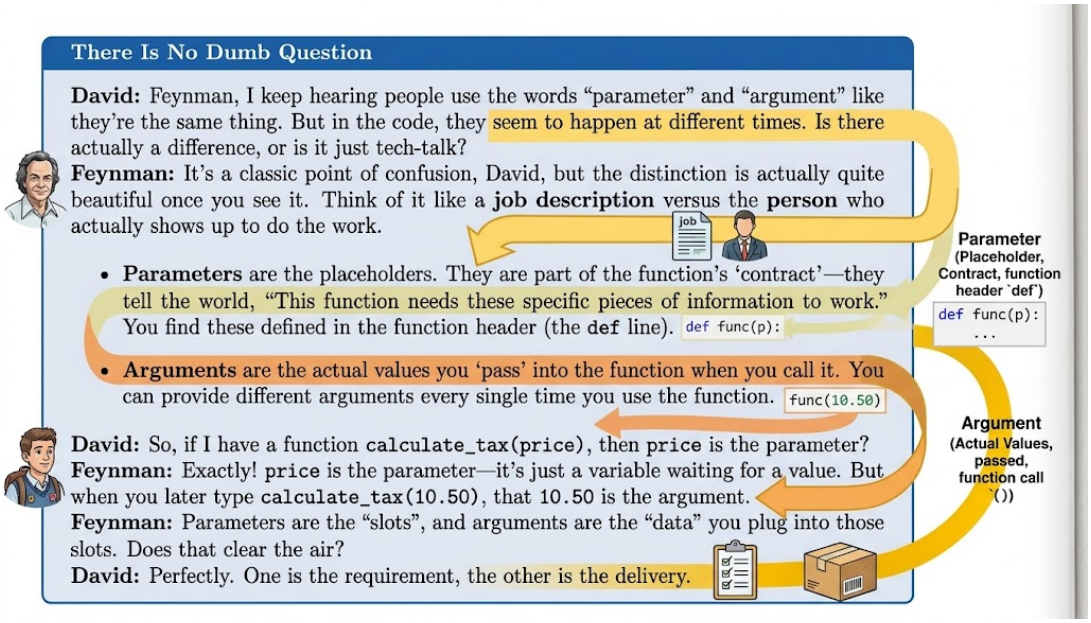


Figure 2: Example of teacher–student dialogue from the Introduction to Programming with Python textbook.

The example above illustrates how dialogue can be used to explain a concept through questions and answers. Authors should aim for natural conversations in which the explanation is clear, accurate, and appropriate for the intended reader.

4.2 Using Illustrations and Visual Explanations

Use diagrams, illustrations, annotated examples, and other visuals when they help students understand a concept or process. A visual should clarify something that might otherwise be difficult to understand from text alone.

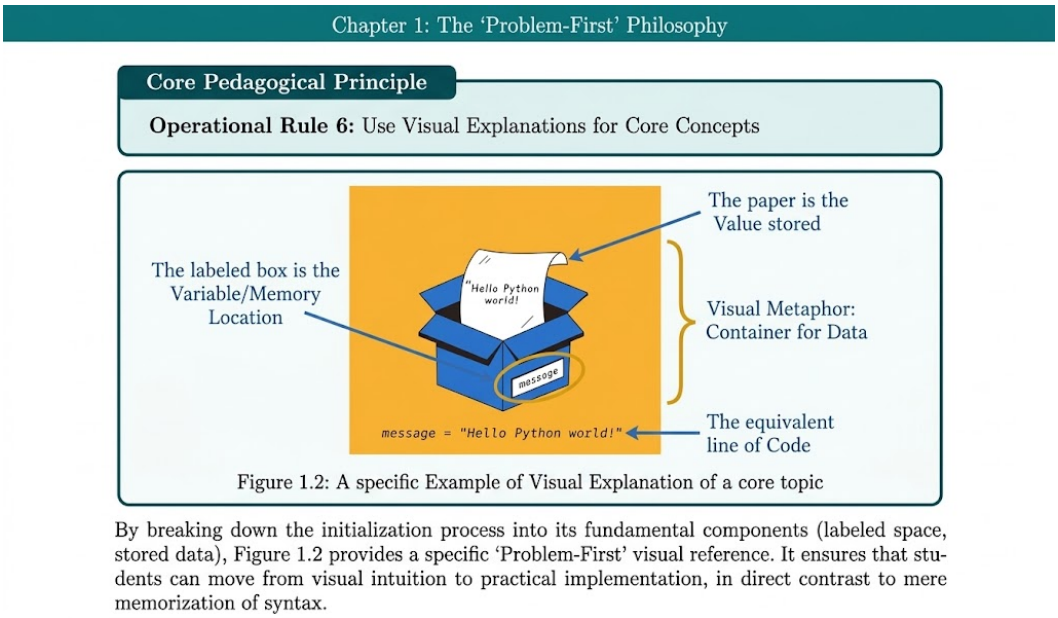


Figure 3: Examples of explanatory visuals used to support learning.

Visuals should be:

- relevant to the concept being taught;
- accurate and consistent with the written explanation;
- clearly labelled where labels are needed; and
- readable at the size at which they will appear.

Authors may use AI-assisted illustrations when appropriate, but they must verify that the resulting images are accurate and useful for learning.

5 How to Develop Lecture Slides

Slides should support the instructor's explanation and help students review the lesson afterwards. They should highlight the important ideas without attempting to replace the full textbook chapter.

Effective slides should:

- present ideas in a clear, logical sequence;
- emphasize important concepts, formulas, diagrams, and code examples;
- use readable text and avoid overcrowding;
- include visuals when they improve understanding; and
- remain useful as a reference after the lesson.

The instructor should provide the detailed explanations, demonstrations, and reasoning that concise slides cannot fully contain.

6 How to Develop Problem Sets

Problem sets should help students test their understanding and develop independent problem-solving skills.

Authors should ensure that:

- each problem clearly states what the student must do;
- the problems match the concepts taught in the course;
- students must reason and apply ideas, not only memorize;
- some problems require applying concepts in unfamiliar situations;
- difficulty increases appropriately where suitable; and
- submission requirements and permitted resources are clearly stated.

Problems should not consist entirely of exercises that repeat worked examples with only minor changes. Students should have opportunities to transfer what they have learned to new tasks.

7 Final Review Checklist

Before teaching a lesson or publishing a learning resource, review the following questions.

7.1 For Every Resource

☐ Is the learning objective clear?

☐ Are the explanations accurate and easy to follow?

☐ Are concepts introduced in a logical order?

☐ Are examples, code, formulas, and visuals correct?

☐ Does every included element serve a learning purpose?

7.2 For Chapters and Lessons

☐ Is there a clear purpose for learning the concept?

☐ Is the reasoning behind important steps explained?

☐ Are examples and visuals used where they help?

☐ Are likely questions and misunderstandings addressed?

7.3 For Problem Sets

☐ Do the problems test understanding and application?

☐ Can students attempt them independently?

☐ Do they require more than copying worked examples?

☐ Are instructions and requirements unambiguous?

8 Reference and Further Reading

The following resources inform and support our approach to teaching and developing learning materials.

- **Introduction to Programming with Python.** CS Made Easy. https://archive.org/details/introduction-to-programming-with-python/page/n41/mode/2up
- **CS Made Easy Textbook Repository.** Source materials and learning resources. https://github.com/nethen-ui/CS-Made-Easy-Textbook/
- **CS50, Harvard University.** A reference for engaging, problem-driven computer science teaching and course design. https://cs50.harvard.edu/

Instructors and material developers are encouraged to consult these resources when preparing lessons and developing learning materials. CS50 is included as a pedagogical reference; the CS Made Easy materials remain the responsibility of CS Made Easy.

9 Our Commitment

Every CS Made Easy instructor and material developer contributes to a shared goal: helping students understand ideas, apply them meaningfully, and become independent problem-solvers.

Teach for understanding.
Start with problems.
Explain through examples.
Develop independent thinking.

Approved by:

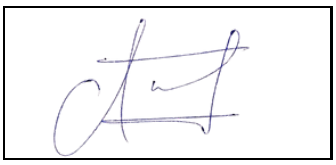
Mikiyas Shegaw, President

Nathan Wondim, Vice President

Date: 23 October 2026



Mikiyas Shegaw
Signature



Nathan Wondim
Signature